CLASS ACTIVITY-20.07.26

NAME : S.PADMAJA

REG NO:192424159

Case Study 1: Hospital Patient Admission Analysis

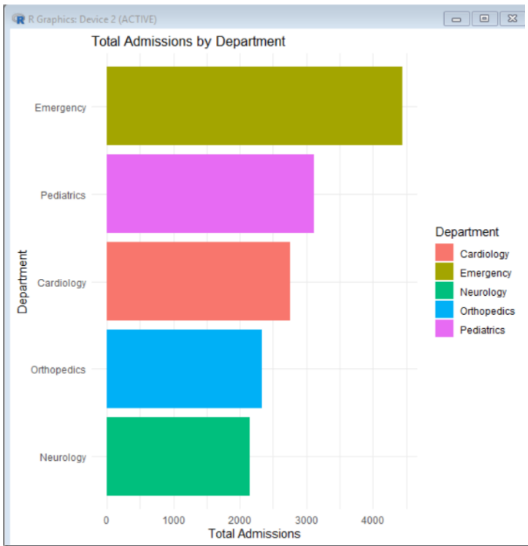
CODE : # Load libraries

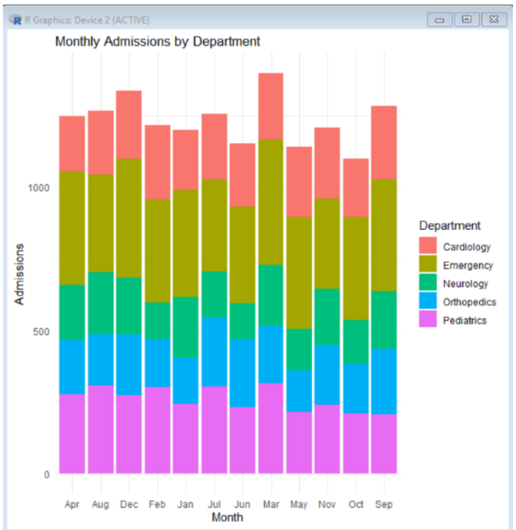
```
library(ggplot2)

library(dplyr)

library(treemap)

set.seed(123)

hospital <- data.frame(

  Department = rep(c("Cardiology", "Neurology", "Pediatrics", "Orthopedics", "Emergency"),
each = 12),

  Month = rep(month.abb, 5),

  Admissions = c(

    sample(180:260, 12),   # Cardiology

    sample(120:220, 12),   # Neurology

    sample(200:320, 12),   # Pediatrics

    sample(140:240, 12),   # Orthopedics

    sample(300:450, 12)    # Emergency

  )

)


# -----------------------------

# 1. Bar Chart - Total Admissions by Department

# -----------------------------

dept_total <- hospital %>%

  group_by(Department) %>%

  summarise(Total_Admissions = sum(Admissions))
```

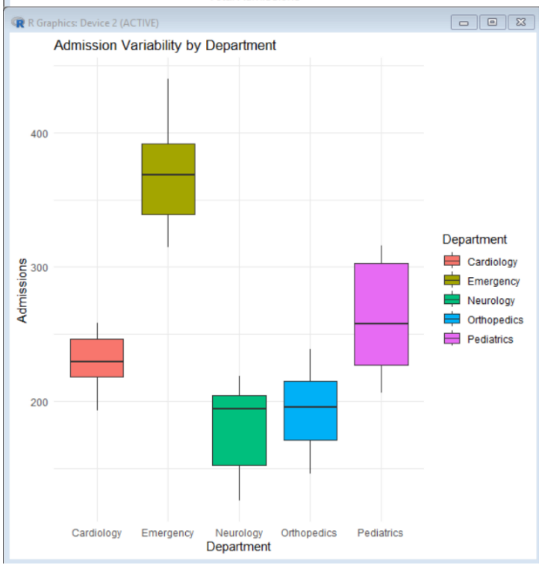
```r
ggplot(dept_total,
       aes(x = reorder(Department, Total_Admissions),
           y = Total_Admissions,
           fill = Department)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  labs(title = "Total Admissions by Department",
       x = "Department",
       y = "Total Admissions") +
  theme_minimal()

# -----------------------------
# 2. Stacked Bar Chart - Monthly Admissions
# -----------------------------
ggplot(hospital,
       aes(x = Month,
           y = Admissions,
           fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Monthly Admissions by Department",
       x = "Month",
       y = "Admissions") +
  theme_minimal()

# -----------------------------
# 3. Treemap - Share of Admissions
# -----------------------------
```

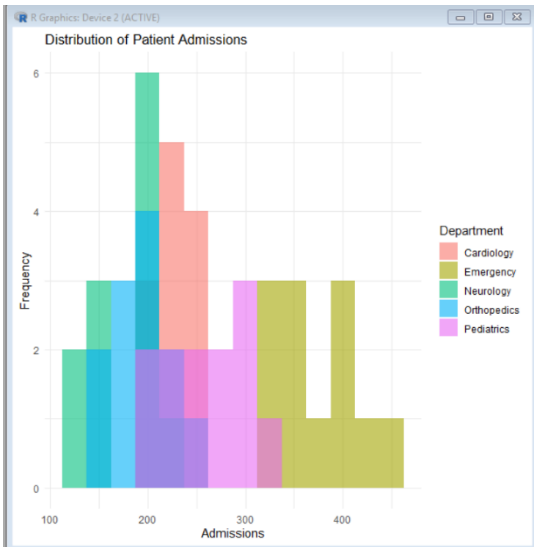
```
treemap(dept_total,
        index = "Department",
        vSize = "Total_Admissions",
        title = "Admission Share by Department")

# -----------------------------
# 4. Box Plot - Admission Variability
# -----------------------------
ggplot(hospital,
       aes(x = Department,
           y = Admissions,
           fill = Department)) +
  geom_boxplot() +
  labs(title = "Admission Variability by Department",
       x = "Department",
       y = "Admissions") +
  theme_minimal()

# -----------------------------
# 5. Histogram - Distribution of Admissions
# -----------------------------
ggplot(hospital,
       aes(x = Admissions,
           fill = Department)) +
  geom_histogram(binwidth = 25,
                 alpha = 0.6,
                 position = "identity") +
  labs(title = "Distribution of Patient Admissions",
```

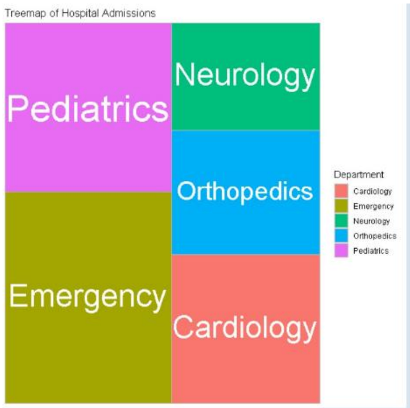
```
    x = "Admissions",
    y = "Frequency") +
  theme_minimal()
```











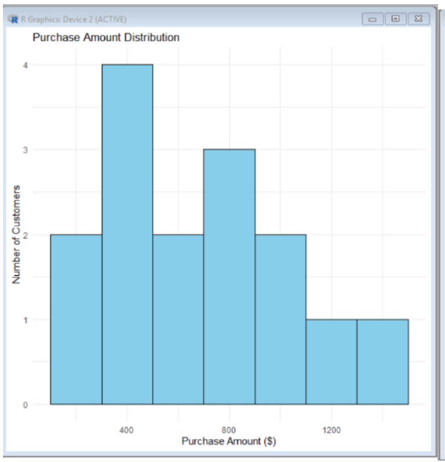
Case Study 2: E-Commerce Customer Spending Behaviour

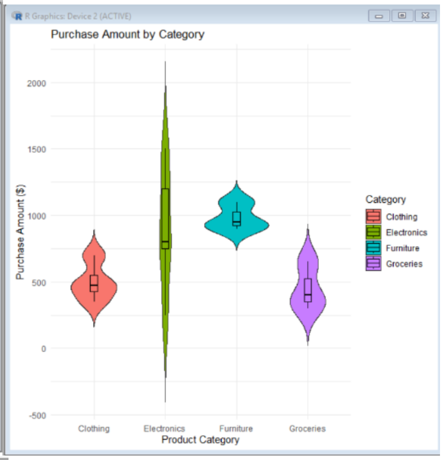
```
# Load library
library(ggplot2)

# Dataset
customer <- data.frame(
  CustomerID = paste0("C", 1:15),
  PurchaseAmount = c(250,500,750,300,1200,900,450,650,1100,350,800,950,400,1500,700),
  OrderFrequency = c(2,5,7,3,10,8,4,6,9,3,7,8,4,12,6),
  Category = c("Electronics","Clothing","Electronics","Groceries",
               "Electronics","Furniture","Clothing","Groceries",
               "Furniture","Clothing","Electronics","Furniture",
               "Groceries","Electronics","Clothing")
)

#-------------------------------------------------
# 1. Histogram
#-------------------------------------------------

ggplot(customer, aes(PurchaseAmount))+
  geom_histogram(binwidth=200,
                 fill="skyblue",
                 color="black")+
  labs(
    title="Purchase Amount Distribution",
    x="Purchase Amount ($)",
    y="Number of Customers"
  )+
```

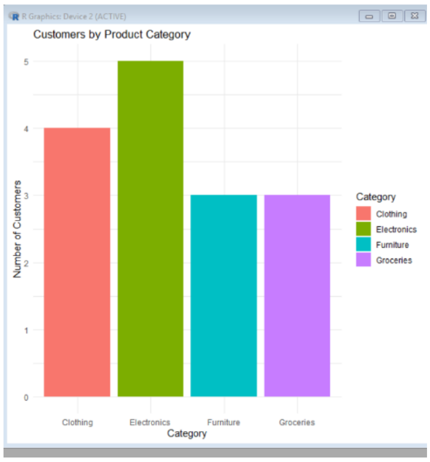
```
  theme_minimal()

#------------------------------------------------
# 2. Violin Plot
#------------------------------------------------

ggplot(customer,
       aes(Category,
           PurchaseAmount,
           fill=Category))+
  geom_violin(trim=FALSE)+
  geom_boxplot(width=0.1,
               color="black")+
  labs(
    title="Purchase Amount by Category",
    x="Product Category",
    y="Purchase Amount ($)"
  )+
  theme_minimal()

#------------------------------------------------
# 3. Box Plot
#------------------------------------------------

ggplot(customer,
       aes(Category,
           PurchaseAmount,
           fill=Category))+
```

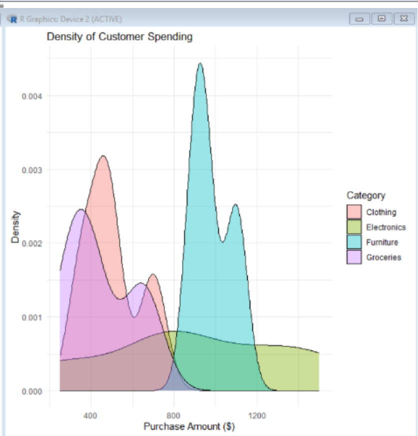
```
  geom_boxplot()+
  labs(
    title="Customer Spending Variation",
    x="Category",
    y="Purchase Amount ($)"
  )+
  theme_minimal()

#------------------------------------------------
# 4. Density Plot
#------------------------------------------------

ggplot(customer,
       aes(PurchaseAmount,
           fill=Category))+
  geom_density(alpha=0.4)+
  labs(
    title="Density of Customer Spending",
    x="Purchase Amount ($)",
    y="Density"
  )+
  theme_minimal()

#------------------------------------------------
# 5. Bar Chart
#------------------------------------------------

ggplot(customer,
```

```
    aes(Category,
        fill=Category))+
geom_bar()+
labs(
  title="Customers by Product Category",
  x="Category",
  y="Number of Customers"
)+
theme_minimal()
```









Case Study 3: University Examination Performance Dashboard

CODE : # Load Library

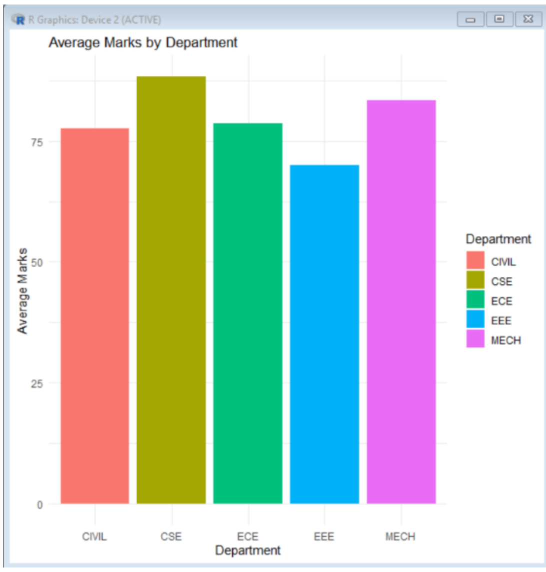
library(ggplot2)

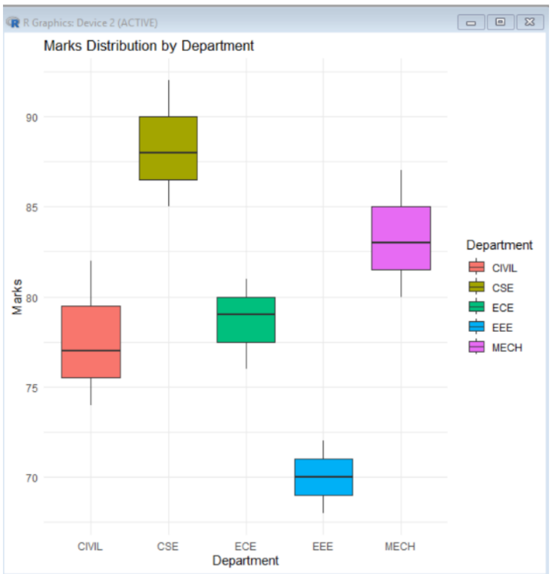
```
# Dataset
student <- data.frame(
  StudentID = paste0("S",1:15),
  Department = c("CSE","CSE","CSE",
                 "ECE","ECE","ECE",
                 "EEE","EEE","EEE",
                 "MECH","MECH","MECH",
                 "CIVIL","CIVIL","CIVIL"),
  Marks = c(92,85,88,
            76,81,79,
            68,72,70,
            83,87,80,
            74,77,82)
)
avg_marks <- aggregate(Marks~Department,
                       student,
                       mean)


ggplot(avg_marks,
       aes(Department,
           Marks,
           fill=Department))+
geom_bar(stat="identity")+
labs(title="Average Marks by Department",
     x="Department",
```

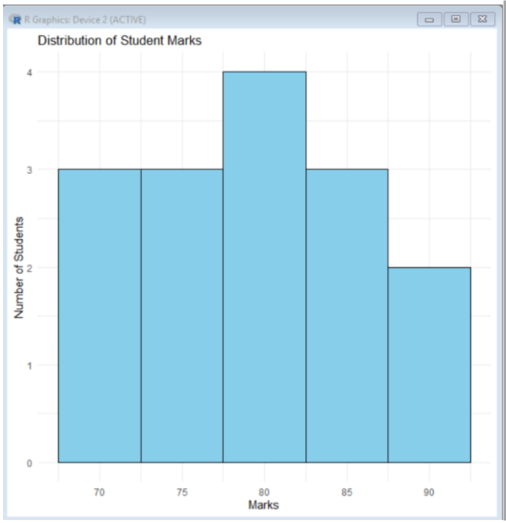
```
    y="Average Marks")+
theme_minimal()

#-----------------------------------------------
# 2. Box Plot
#-----------------------------------------------

ggplot(student,
      aes(Department,
          Marks,
          fill=Department))+
geom_boxplot()+
labs(title="Marks Distribution by Department",
    x="Department",
    y="Marks")+
theme_minimal()

#-----------------------------------------------
# 3. Histogram
#-----------------------------------------------

ggplot(student,
      aes(Marks))+
geom_histogram(binwidth=5,
              fill="skyblue",
```

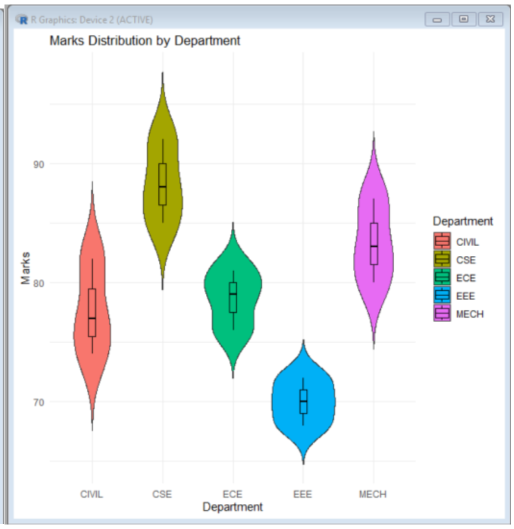
```
            color="black")+
labs(title="Distribution of Student Marks",
    x="Marks",
    y="Number of Students")+
theme_minimal()

#-----------------------------------------------
# 4. Violin Plot
#-----------------------------------------------

ggplot(student,
      aes(Department,
          Marks,
          fill=Department))+
geom_violin(trim=FALSE)+
geom_boxplot(width=0.1,
           color="black")+
labs(title="Marks Distribution by Department",
    x="Department",
    y="Marks")+
theme_minimal()

#-----------------------------------------------
# 5. Scatter Plot
#-----------------------------------------------
```

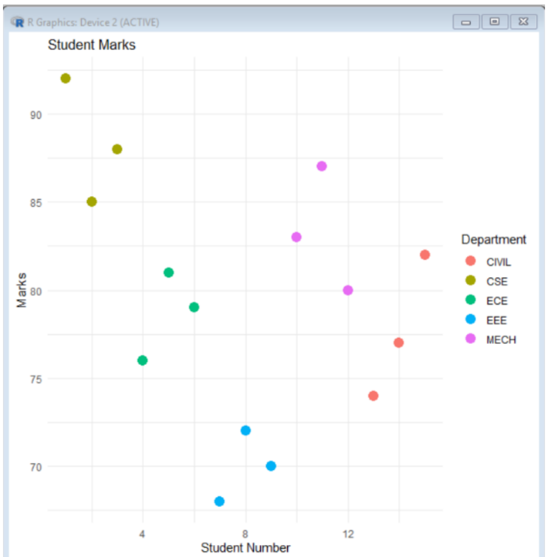
```
student$StudentNo <- 1:nrow(student)

ggplot(student,
       aes(StudentNo,
          Marks,
          color=Department))+
geom_point(size=4)+
labs(title="Student Marks",
     x="Student Number",
     y="Marks")+
theme_minimal()
```











Case Study 4: Climate and Rainfall Distribution Study

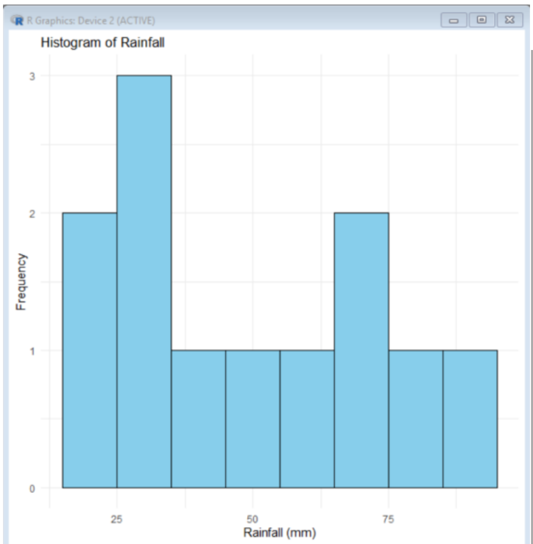
CODE :

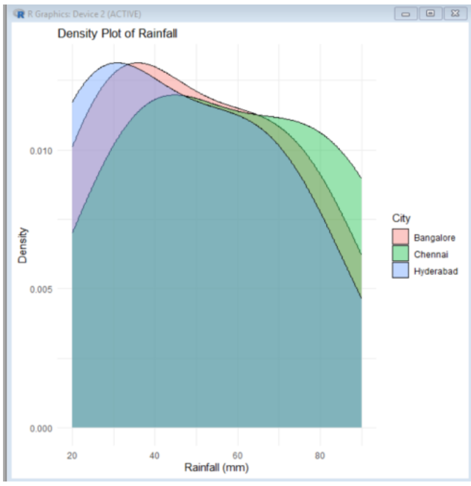
```
# Load Library

library(ggplot2)


# Dataset

climate <- data.frame(

  City = c("Chennai","Bangalore","Hyderabad",

           "Chennai","Bangalore","Hyderabad",

           "Chennai","Bangalore","Hyderabad",
```

```
        "Chennai","Bangalore","Hyderabad"),

  Month = c("Jan","Jan","Jan",
            "Feb","Feb","Feb",
            "Mar","Mar","Mar",
            "Apr","Apr","Apr"),

  Rainfall = c(30,25,20,
               45,35,30,
               70,60,55,
               90,80,75),

  Temperature = c(29,25,28,
                  30,26,29,
                  32,28,31,
                  34,30,33),

  Humidity = c(70,65,60,
               72,68,63,
               78,72,69,
               82,76,73)
)

#------------------------------------------------
# 1. Histogram
#------------------------------------------------

ggplot(climate,
```

```
      aes(Rainfall))+
geom_histogram(binwidth=10,
             fill="skyblue",
             color="black")+
labs(title="Histogram of Rainfall",
    x="Rainfall (mm)",
    y="Frequency")+
theme_minimal()

#-----------------------------------------------
# 2. Density Plot
#-----------------------------------------------

ggplot(climate,
      aes(Rainfall,
          fill=City))+
geom_density(alpha=0.4)+
labs(title="Density Plot of Rainfall",
    x="Rainfall (mm)",
    y="Density")+
theme_minimal()

#-----------------------------------------------
# 3. Box Plot
#-----------------------------------------------

ggplot(climate,
      aes(City,
```

```r
          Rainfall,
          fill=City))+
geom_boxplot()+
labs(title="Rainfall Distribution by City",
    x="City",
    y="Rainfall (mm)")+
theme_minimal()

#-----------------------------------------------
# 4. Bar Chart
#-----------------------------------------------

ggplot(climate,
      aes(Month,
          Rainfall,
          fill=City))+
geom_bar(stat="identity",
        position="dodge")+
labs(title="Monthly Rainfall by City",
    x="Month",
    y="Rainfall (mm)")+
theme_minimal()
```





Case Study 5: Bank Loan Risk Assessment

CODE :

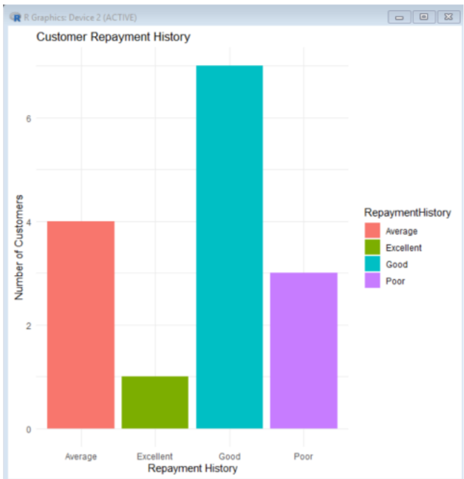
```
# Load Library

library(ggplot2)


# Dataset

loan <- data.frame(

  CustomerID = paste0("C", 1:15),

  LoanAmount = c(50000,120000,80000,150000,90000,

               70000,200000,110000,95000,130000,

               175000,85000,100000,60000,140000),
```

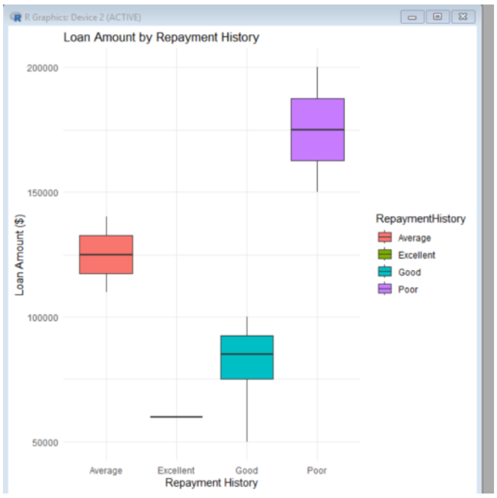
```
  Income = c(40000,75000,55000,90000,60000,
             50000,120000,70000,65000,80000,
             100000,58000,72000,45000,85000),

  CreditScore = c(720,680,750,640,700,
                  730,610,690,710,670,
                  620,740,705,760,650),

  RepaymentHistory = c("Good","Average","Good","Poor","Good",
                       "Good","Poor","Average","Good","Average",
                       "Poor","Good","Good","Excellent","Average")
)

#------------------------------------------------
# 1. Histogram - Loan Amount
#------------------------------------------------

ggplot(loan, aes(LoanAmount)) +
  geom_histogram(binwidth = 25000,
                 fill = "skyblue",
                 color = "black") +
  labs(
    title = "Distribution of Loan Amount",
    x = "Loan Amount ($)",
    y = "Number of Customers"
  ) +
  theme_minimal()
```

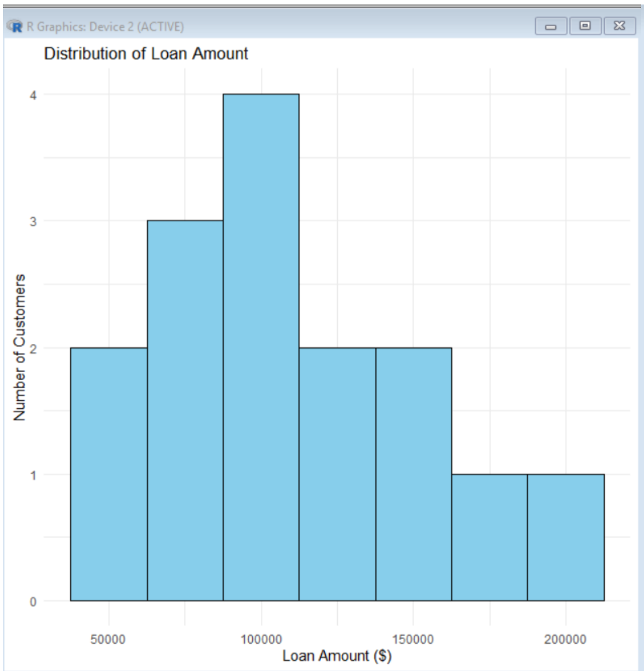
```r
#-----------------------------------------------

# 2. Box Plot - Loan Amount by Repayment History

#-----------------------------------------------


ggplot(loan,

       aes(RepaymentHistory,

           LoanAmount,

           fill = RepaymentHistory)) +

  geom_boxplot() +

  labs(

    title = "Loan Amount by Repayment History",

    x = "Repayment History",

    y = "Loan Amount ($)"

  ) +

  theme_minimal()


#-----------------------------------------------

# 3. Scatter Plot - Income vs Credit Score

#-----------------------------------------------


ggplot(loan,

       aes(Income,

           CreditScore,

           color = RepaymentHistory,

           size = LoanAmount)) +

  geom_point(alpha = 0.7) +

  labs(
```

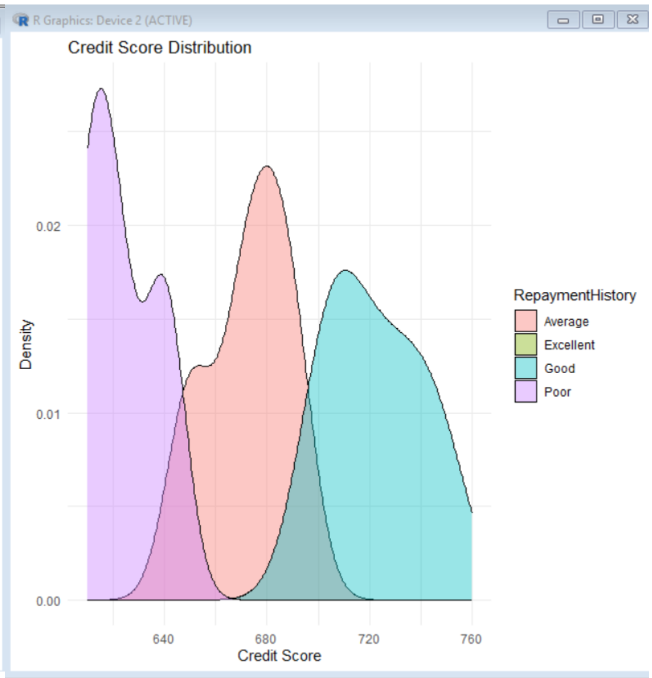
```
    title = "Income vs Credit Score",
    x = "Income ($)",
    y = "Credit Score"
  ) +
  theme_minimal()

#------------------------------------------------
# 4. Bar Chart - Customers by Repayment History
#------------------------------------------------

ggplot(loan,
       aes(RepaymentHistory,
           fill = RepaymentHistory)) +
  geom_bar() +
  labs(
    title = "Customer Repayment History",
    x = "Repayment History",
    y = "Number of Customers"
  ) +
  theme_minimal()

#------------------------------------------------
# 5. Density Plot - Credit Score
#------------------------------------------------

ggplot(loan,
       aes(CreditScore,
           fill = RepaymentHistory)) +
```

```
geom_density(alpha = 0.4) +
labs(
  title = "Credit Score Distribution",
  x = "Credit Score",
  y = "Density"
) +
theme_minimal()
```






R Graphics: Device 2 (ACTIVE)
Distribution of Loan Amount
Number of Customers
Loan Amount ($)
0
1
2
3
4
50000
100000
150000
200000


R Graphics: Device 2 (ACTIVE)
Credit Score Distribution
Density
Credit Score
0.00
0.01
0.02
640
680
720
760
RepaymentHistory
Average
Excellent
Good
Poor